Diabetes Prediction

1. Problem Understanding

Description of the Problem

Diabetes is a chronic health condition that affects how the body regulates blood glucose levels. Early identification of individuals at risk is critical to prevent complications such as cardiovascular disease, kidney failure, and nerve damage.

This project aims to build predictive models that can classify whether an individual is likely to have diabetes based on health-related features such as age, BMI, blood pressure, smoking status, and exercise factors.

Importance of Predicting Diabetes Risk

Accurate prediction of diabetes risk enables:

- **Early intervention**, reducing long-term complications
- **Better healthcare planning** and resource allocation
- **Personalised treatment strategies** for high-risk individuals

From a machine learning/deep learning perspective, this is a **binary classification problem**, where the goal is to distinguish between diabetic and non-diabetic individuals.

2. Data Exploration Insights

Key Observations from the Dataset

- The dataset is **highly imbalanced**, with significantly more non-diabetic cases than diabetic cases (approximately 9:1 ratio).
- Features such as **BMI, age, and blood pressure** show noticeable variation between diabetic and non-diabetic groups.
- Some features exhibit **skewness and outliers**, particularly BMI and blood pressure.
- Strong **correlation and multicollinearity** exist among certain features, which can negatively impact linear models.

Important Features Influencing Prediction

- **BMI**: Higher BMI values are generally associated with increased diabetes risk.
- **Age**: Older individuals tend to have a higher likelihood of diabetes.
- **Blood Pressure**: Elevated levels may indicate underlying health risks.

- **Smoking Status**: Lifestyle factors such as smoking contribute to risk.

To address multicollinearity in Logistic Regression, **Principal Component Analysis (PCA)** was applied to transform correlated features into independent components.

Machine Learning Model (Logistic Regression)

Logistic Regression was used as a baseline model due to its simplicity and interpretability. Since multicollinearity was present in the dataset, **PCA was applied before training** to convert correlated variables into orthogonal principal components.

Key aspects:

- Input features transformed using PCA
- Class imbalance handled using **class weighting**
- Model outputs probability of diabetes risk

Neural Network Architecture

A feedforward neural network was implemented to capture potential non-linear relationships in the data.

Architecture:

- Input layer corresponding to original features (no PCA applied)
- One or more hidden layers with **ReLU activation**
- **Dropout layers (0.1–0.4)** to reduce overfitting
- **L2 regularisation** applied to weights
- Output layer with **sigmoid activation** for binary classification

Training strategies:

- **Class weighting** to address imbalance
- **ReduceLROnPlateau** for adaptive learning rate
- **Early stopping** to prevent unnecessary training

4. Model Evaluation

Performance Metrics

Since the dataset is highly imbalanced, multiple metrics were used instead of relying only on accuracy.

Logistic Regression Results

- **Accuracy:** 65.1%

- **Precision:** 0.568
- **Recall:** 0.682
- **F1 Score:** 0.529

The model performs well in terms of **recall**, meaning it can detect most diabetic cases, though it produces some false positives.

Neural Network Results

- **Accuracy:** 60.47%
- **Precision:** 0.553
- **Recall:** 0.661
- **F1 Score:** 0.489
- **AUC:** 0.74

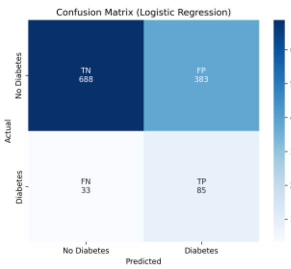
The neural network shows **reasonable classification ability (AUC)** but slightly lower performance across all metrics compared to Logistic Regression.

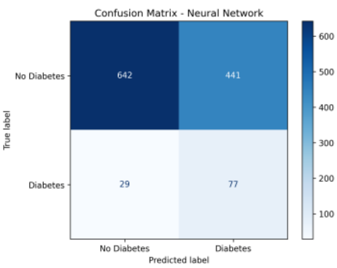
Overall Interpretation

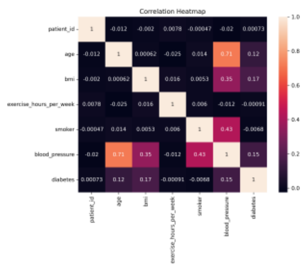
- Logistic Regression performs **better overall**, especially in recall and F1 score
- Neural Network has **good AUC**, but does not outperform the simpler model
- Both models are **recall-focused**, which is important for detecting diabetic cases

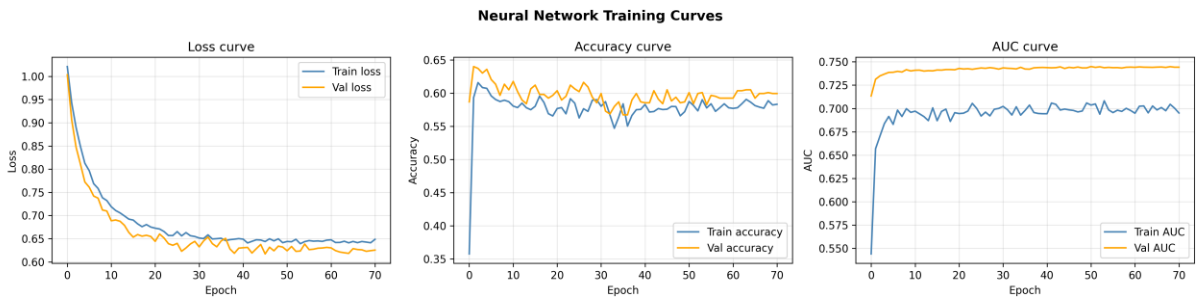
Visualizations

Several visual tools were used to evaluate model performance









Comparative Analysis

Differences Between the Models

Aspect	**Logistic Regression**	**Neural Network**
Complexity	Simple	Complex
Interpretability	High	Low
Feature Handling	PCA applied	Raw features
Training Time	Fast	Slower
Overfitting Risk	Low	Higher (controlled with regularisation)

Reasons for Performance Differences

- The dataset is **small and low-dimensional**, limiting the advantage of neural networks
- Logistic Regression benefits from **PCA**, which removes multicollinearity
- Neural networks require larger datasets to fully capture complex patterns
- Strong regularisation in the neural network leads to **slight underfitting**

Recommendations for Real-World Use

- **Logistic Regression is preferred** for this problem due to:
  - Comparable performance
  - Better interpretability
  - Lower computational cost
- Neural networks may be considered if:
  - More data becomes available
  - Additional complex features are introduced